DE–4161 **101**

DISTANCE EDUCATION

M.C.A. (Semester) DEGREE EXAMINATION, MAY 2012.

First Semester

DIGITAL COMPUTER ORGANISATION

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Convert the hexadecimal number $(ABF \cdot 12)_{16}$ to decimal and binary.

2. Using De Morgan's law, write an expression for the complement of F if $F(x, y, z) = xy + \overline{x}z + y\overline{z}$.

3. Explain the circuit and working of a demultiplexer.

4. What is ALU? Draw the block diagram of a 4 bit ALU.

5. What is ROM? Explain its types.

6. Minimize the following using Karnaugh maps :

   $Y = \overline{D}\,\overline{C}\,\overline{B} + D\overline{C}\,\overline{B} + C\overline{B}A + \overline{C}B\overline{A}$.

7. What are NOR and NAND gates? Why are they called universal gates? Explain.

8. What are interrupt? Explain what happens when a computer recognizes an interrupt.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. What are the basic laws of Boolean algebra? Explain.

10. Discuss the working of a parallel binary adder and subtractor with suitable diagram.

11. Explain J-K master slave flip-flop with neat diagram.

12. Explain in detail about up-counter and down-counter with suitable diagram.

13. Discuss the different phases of instruction execution cycle.

14. Explain in detail about program controlled I/O.

15. Explain in detail the DMA data transfer with a schematic diagram.

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DE–4162 **102**

DISTANCE EDUCATION

M.C.A.(Semester) DEGREE EXAMINATION, MAY 2012.

First Semester

C AND DATA STRUCTURES

(2010 Academic year Onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Explain the break statement in C by give an example.

2. Write a note on character set in C.

3. Explain briefly about Macros.

4. Write a note on library function with an example.

5. Explain how to processing a data file.

6. Discuss about the applications of Queue.

7. Write a C program to reverse a string manually.

8. Explain briefly about Binary Tree representation.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about any three branching statements. (15)

10. Write a C program that appends one file at the end of another. (15)

11. Explain briefly about:

    (a) Passing an array to functions. (8)

    (b) Passing arguments to a function. (7)

12. Explain how to create and processing a data file. (15)

13. Write a C program to find addition two matrices. (15)

14. Explain in detail about Binary Tree Traversal. (15)

15. Write a C program to generate a marksheet using file handling. (15)

DE–4163 **103**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION,
MAY 2012.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE of the following questions.

1. Discuss the advantages and disadvantages of DBMS.

2. How data can be manipulabed in SQL? Explain.

3. Define SQL. What are the advantages and disadvantages of SQL?

4. List and explain the important rules for sub queries.

5. Write down the syntax for creating a table, inserting record into table and retrieving the records from table.

6. What do you mean by constraints? Explain with examples.

7. What is meant by record? Discuss its benefits.

8. List and explain the commands available in DML.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR of the following questions.

9. Outline the advantages of implementing database management system in an organization.

10. What do you mean by structure of a relational model of a database system? Explain the significance of domain and keys in the relational model.

11. Discuss in detail about oracle database architecture.

12. Describe the basic features of the SQL. Discuss their advantages disadvantages and importance to the end user and the designer.

13. Write a brief note on embedded SQL statements.

14. Define foreign keys. What is this concept used for? How does it play a role in the join operation?

15. Explain briefly various types of database with suitable illustrations.

DE–4164 **104**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

First Semester

OPERATING SYSTEMS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the structure of an operating system.

2. Briefly explain the inter process communication.

3. Write a note on Semaphores.

4. With neat diagram, explain the interrupt handler procedures.

5. Briefly explain Banker's algorithm with example.

6. Explain the concept of multiprogramming.

7. Discuss about segmentation in memory management.

8. Write about the various file operations.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe various types of operating systems.

10. Compare and contrast between priority scheduling and round robin scheduling.

11. Describe the deadlock prevention and avoidance techniques.

12. Explain the first - fit strategy used in multiple partition allocation with an example.

13. Describe the file system with organization in detail.

14. Explain the various process states with a neat diagram.

15. Write short notes on the following:

    (a) servers (5)

    (b) disk drivers (5)

    (c) virtual memory (5)

DE–4165 **201**

DISTANCE EDUCATION

M.C.A. (Semester) DEGREE EXAMINATION, MAY 2012.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Differentiate between procedure oriented and object oriented programming.

2. What are the operators in C++ program?

3. Explain 'if' structure with an example.

4. Describe briefly about "call by reference" with an example.

5. What is inline function with an example?

6. Explain the 'Destructor' with an suitable example.

7. Explain briefly 'types conversion'.

8. How will you access command line argument? Explain it with an example.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain benefits of object oriented programming in detail.

10. (a) What is pointer? What are its benefits? (7)

(b) What is an array? Explain array of object with an suitable example. (8)

11. What is friend function? Explain with an suitable example.

12. Explain with suitable example of binary operator overloading.

13. What is inheritance? Explain briefly different forms of inheritance.

14. Write a short notes on:

(a) 'This' pointer

(b) Standard manipulator.

15. Explain the file processing features of C++.

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DE–4166 **202**

DISTANCE EDUCATION

M.C.A. (Semester) DEGREE EXAMINATION, MAY 2012.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define Directory. Describe directory hierarchy.

2. Give short notes on Inodes.

3. What is conditional expression? Explain with an example.

4. Describe various options of shell attributes.

5. What is the purpose of 'cal' command?

6. Describe the 'zap' and 'pick' commands.

7. What is file system? Explain.

8. Write a short note on MS Macro package.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain the features of awk pattern learning and processing language.

10. Explain the various loop statements in shell programming with an example.

11. Write a shell programming to find out sum of 'N' odd numbers.

12. Briefly explain the grep family in unix.

13. Explain various control statements in shell programming.

14. What is function? How is it differ from procedure? Explain.

15. Write a short note on the following shell commands.

    (a) sed

    (b) tbl and eqn

    (c) diff

    (d) tail

    (e) sort.

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DE–4167 **203**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

Second Semester

COMMUNICATION SKILLS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the process of communication.

2. Comment on the various reading styles.

3. What are the listening tests?

4. Analyse the interview techniques.

5. Describe the tool for personality identification.

6. What are the negotiation techniques?

7. Write short note on the importance of mock mutings.

8. Discuss communication within management.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain the commandments of communication.

10. How will you improve your english language writing?

11. What are the methods of asking questions?

12. Discuss the process of group discussion.

13. Write an essay on interpersonal behaviour.

14. Discuss downward and upward communication.

15. What are the techniques for delivering presentation?

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DE–4168 **204**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

Second Semester

COMPUTER GRAPHICS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. List the applications of computer graphics.

2. (a) Illustrate the difference between random scan systems and raster scan systems.

   (b) Explain the direct view storage tube.

3. Describe the Bresenham's line drawing algorithm.

4. Explain the two dimensional transformations.

5. Discuss the midpoint subdivision line clipping algorithm.

6. Explain any three 3-D display techniques.

7. Describe the 3-D composite transformations.

8. Write a note on command language.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the ellipse generation algorithm.

10. Describe the region filling techniques.

11. Discuss the 2D matrix representation and 2D composite transformations.

12. Explain the concept of polygen clipping.

13. Explain the parallel projection and perspective projection.

14. Describe the hidden surface removal algorithms.

15. List the components of user interface.

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DE–4401 **105**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2012.

First Semester

DATA STRUCTURE USING C LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select One question by the lot system and give it to the student.

1. (a) Write C program to implement push and pop operation of stack. (50)

   (b) Write a C program to find a given word is palindrome or not. (40)

   Record. (10)

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2. (a) Define selection sort and write a C program to sort a set of elements using selection sort. (50)

   (b) Write a C program to find the factorial of a given number using recursion. (40)

   Record. (10)

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3. (a) Write a C program to evaluate any post fix mathematical expression using stack. (50)

   (b) Write a C program to copy contents of one file to another. (40)

   Record. (10)

4. (a) Write a C program for implementing queue ADT with necessary operations using array. (50)

(b) Write C program to find the roots of a quadratic equation using function definition. (40)

Record. (10)

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5. (a) Write a C program to sort the set of numbers in ascending order with naming of variable and the value before and after sorting. (45)

(b) Write a C program to read 10 values to an array variable. Use pointers to locate and display each value. (45)

Record. (10)

DE–4402 **106**

DISTANCE EDUCATION

M.C.A. (S) (Computer Applications) DEGREE EXAMINATION, MAY 2012.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the lot system and give it to the student.

1. (a) Create a table EMPLOYEE to store the following information :

Empcode

Employeename

Designation

DOB

Sex

Basic pay

Write query to do the following :

(i) Display empcode and designation of all the employees

(ii) Display the names of female employee who are getting basic pay greater than 6000.

(iii) Display the name and designation of employees having age greater than 50

(iv) Add a new column department no in the above table anyd insert values in that column

(v) Give a 5% increment to the basic pay of employees whose designation is 'Manager'.

(b) Consider the following database of student enrollment in courses and books adopted for each course.

STUDENT (regno, name, major, bdate)

COURSE (course#, cname,dept)

ENROLL (regno, course, sem, marks)

BOOK_ADOPTION (course, sem, book-ISBN)

TEXT (book-ISBN, book-title, publisher, author)

(i) Create the above table by properly specifying the primary key and the foreign keys

(ii) Enter atleast five tuples for each relation

(iii) Demonstrate how you add a new text book to the database and make this book be adopted by some department

(iv) Produce a list of text books (include course#, Book-ISBN, book-title) in the alphabetical order for courses offered by the 'CS' department that use more than two books

(v) List any department that has all its adopted books published by a specific publisher.

2. (a) Consider the Insurance database given below. The primary key are underlined and the data type are specified.

PERSON (driver-id, name, address)

CAR (Regno, model, year)

ACCIDENT (report-number, acc-date, location)

OWN (driver-id#, Regno)

PARTICIPATED (driver-id#, Regno, report-number, damage-amount)

(i) Create the above tables by properly specifying the primary keys and the foreign keys

(ii) Enter atleast five tuples for each relation

(iii) Find the total number of people who owned cars that were involved in accidents in 2002

(iv) Find the number of accidents in which cars belonging to a specific model were involved

(v) Generation of suitable reports.

(b) Create the following tables :

Student (roll-no, name, date-of-birth, course-id) course (course-id, name, fee, duration)

Write a query to do the following :

(i) List all those students who are greater than 18 years of age and have opted for MCA course

(ii) List all those courses whose fee is greater than that of MCA course

(iii) Find the number of students who have enrolled for the subject "DBMS".

3. (a) Create the following table named customer :

Customer id
Name
Area
Phone

Insert the appropriate data into table :

(i) Update phone numbers of all customer to have a prefix as your city STD code

(ii) Print the entire customer table

(iii) List the names of those customer who have 'e' a second letter in their names

(iv) Display all records in increasing order of name

(v) Display all those records from customer table where name starts with 'a' or area is "abc"

(vi) Display all records of those where name starts with 'a' and phone exchange is 55.

(b) Create a table of employee (emp_number, name, dept, salary) and department (dept number, dept name) Insert some records in the table through appropriate forms having integrity checks. Add some records in employee table where department value is not present in department table.

(i) Display all records from employee table where department is not found in department table

(ii) Display records from employee table in a report format with proper headings. This report must also contain those records where department number does not match with any value of department table

(iii) Display those employee records who have salary less than the salary of person whose empcode = 'A100'

(iv) Create another table : Sales_data (Region-code, city, sales person-code, sales-qty)

(v) Display records where sales person has achieved sale more than average sales of all sales persons of all the regions.

4. (a) Create the following tables :

Student (roll-no, name, category, district, state, DOB, course-id)

Student-rank (roll-no, marks, rank)

Course (course-id, name, fee, duration)

Write Query to do the following :

(i) List names of the students who are having same rank but they should reside in different districts

(ii) List details of student they belongs to same category with same rank

(iii) List all those students who are between 18-19 years of age and have opted for MCA course

(iv) List all those courses in which numbers of students are less than 10.

(b) Create the following tables for a University :

Student (St_id, St_name, St_progru-hu, St_dob)

Subjects (Su_id, Su_name, Su_programme)

Student_Subject (St_id, Su_id, mark_obtained)

Write query to do the following :

(i) List the id and name of all the student who have passed a subject having subject id "MCS013"

(ii) List the result of all the students as : St_id, St_name, St_programme, Su_id, Su_name, marks_obtained)

(iii) Find the DOB (date of birth) of the youngest and oldest student of the university

(iv) Find the id and name of the student who have passed at least one subject.

5. (a) A hospital maintains blood donors record information in a file. The file contains the following fields :

Donor number, Donar name, Donar age, Donor address, Pincode, Place of birth, Blood group (A, B, AB & O).

Write a program using Oracle to print out the number, name and address of the donors for the following categories :

(i) Blood donor having AB blood group

(ii) Blood donor in the age group between 16 to 25

(iii) Female donors with 'A' blood group in the age between 18 to 25.

(b) Create a table of employee (emp-number, name, dept, Date of joining, designation, salary) and department (dept number, dept name)

Write a query to do the following :

(i) Display all employees who are working in department 20 or 30 and working as either manager or analyst

(ii) Update all employee salary with the existing salary plus 10% of that salary who have joined before 2006

(iii) Find out the employees whose salary is more than the average salary of their own department

(iv) Delete all those employees whose designation is either clerk or salesman.

DE–4169 **301**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain by giving examples primitive and molecular statements.

2. Show that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$.

3. Define partial order relation. Give an example. Also explain the use of Hasse diagram.

4. Differentiate between function and relations.

5. Explain Hashing functions with examples.

6. Define Semigroups and monoids. Give examples.

7. Define graph, isomorphic graph and directed graph with examples.

8. Define patho, connectedness in graphs with examples.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain the connectives in mathematical logic.

10. Explain the properties of binary relations on a set.

11. Let $R = \{(1,2),(3,4)\ (2,2)\}$ and

$S = \{(4,2), (2,5), (3,1), (1,3)\}$

Find $R * S, S * R, R * (S * R), (R * S) * R, R * R, S * S,$ and $R * R * R$.

12. Explain the various definitions in case of binary and n-ary operations with examples.

13. Explain subsemigroups and submonoids with examples.

14. Explain cosets and Lagrange's theorem.

15. Explain the various aspects of trees with examples.

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DE–4170 **302**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

COMPUTER NETWORKS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List the uses of Computer Networks.

2. Write any one transmission media with neat diagram.

3. Explain sliding window protocol.

4. Write a short note on CSMA protocols.

5. Explain the elements of transport protocols.

6. Write a short note on address resolution protocol.

7. What is private key and public key encryption?

8. Discuss the basic concepts of dual signature.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the OSI reference model with neat diagram.

10. Explain Broadband and Narrowband ISDN.

11. Explain in detail about finite state model.

12. Discuss the distance vector routing and flow based routing in detail.

13. Explain in detail about ATM network.

14. Explain the DES algorithm with a neat block diagram.

15. With a neat block diagram, discuss the architecture of SNMP and its components in detail.

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DE–4171 **303**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) Lateral DEGREE EXAMINATION, MAY 2012.

Third Semester

SOFTWARE ENGINEERING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the process model.

2. Define software Metrics.

3. Explain about COCOMO model.

4. Explain the different techniques of project Cost estimation.

5. Discuss about quality assurance in detail.

6. Explain how to decompose the project for cost estimation.

7. What is software prototyping?

8. Explain about system and validation testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about the various role of management in software development.

10. Explain the project estimation in detail.

11. Discuss about the quality management.

12. Explain about coupling and cohesion in detail.

13. Discuss the different types of testing in detail.

14. Discuss about Interface Design and Guidelines.

15. Explain about project staffing.

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DE–4172 **304**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

VISUAL PROGRAMMING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List eight form properties and explain them.

2. What is programming user events?

3. Explain the grid controls.

4. Explain the menus in VC++.

5. How will you import VBX controls?

6. Explain briefly about splitter window.

7. Write a short notes on dynamic data transfer.

8. Explain the concept of serialization.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Write a VB program to retrieve the marks of the student from database and then compute the rank and total.

10. Explain the inbuilt and user defined functions.

11. Describe about VC++ components.

12. List out mouse and keyboard events.

13. Explain the process of creating user defined DLL'S.

14. Discuss about object Linking and Embedding.

15. Explain the MFC file handling in detail.

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DE–4173 **401**

DISTANCE EDUCATION

M.C.A (S) /M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Journalise the following transactions in the books of Imran:

   (a) Started business with cash Rs. 45,000

   (b) Goods purchased for cash Rs.15,000

   (c) Sold goods for cash Rs.8,500

   (d) Withdrew from bank for private use Rs.1,000

   (e) Paid for stationery Rs.200.

2. Explain the objectives and limitations of ratio analysis.

3. Following are the balance sheet of Paliwal Exports Limited for the year 2010 and 2011.

Liabilities	2010 Rs.	2011 Rs.	Assets	2010 Rs.	2011 Rs.
Creditors for Goods	1,60,000	2,50,000	Stock	2,00,000	2,70,000
Creditors for Expenses	10,000	12,000	Sundry Debtors	2,25,000	2,45,000
Bills Payable	1,00,000	1,00,000	Cash	40,000	65,000

Liabilities	2010 Rs.	2011 Rs.	Assets	2010 Rs.	2011 Rs.
Share Capital	5,50,000	6,20,000	Prepaid Expenses	25,000	22,000
Securities Premium	50,000	80,000	Plant & Machinery	7,00,000	8,80,000
Profit & Loss Account	1,00,000	2,00,000	Goodwill	1,00,000	70,000
Debentures	3,00,000	2,00,000	Long-term Investment	1,80,000	1,80,000
General Reserve	2,00,000	2,60,000			
Total	14,70,000	17,32,000		14,70,000	17,32,000

Prepare cash flow statement.

4. The Indian House Limited is a large retailer in manufacturing equipment. The company's profit and loss account for the year ended 31st March, 2011 is given under:

	Rs.	Rs.
Sales		24,00,000
Less: Cost of goods manufactured (variable)		15,00,000
Gross Margin		9,00,000
Less: Operating Expenses:		
Selling Expenses	3,00,000	
Administration expenses	1,50,000	
Net profit		4,50,000
		4,50,000

The selling expenses are 90% variable and the administration expenses are 20% variable. You are asked to prepare income statement under the marginal costing method.

5. What are the advantages and limitations of standard costing?

6. A manufacturing company submits for the following figures relating to Product X for the first quarter of 2010.

Sales Targets:	January	60,000 units
	February	48,000 units
	March	72,000 units
Stock position:	1st January 2010 (% of January 2010 sales)	50%
	31st March, 2010	40, 000 units
	End of January & February (% of subsequent month sales)	50%

You are required to prepare production budget for the first quarter of 2010.

7. Explain in detail about the functions of financial management.

8. Make a choice of the project on the basis of bailout payback period from the following information.

	Project A	Project B
	Rs.	Rs.
Life	8 years	8 years
Initial investment	10,00,000	15,00,000
Cash inflows per year	2,50,000	4,00,000

Salvage values expected at the end of 1St year are Project A Rs. 6,00,000 declining at an annual rate of Rs.1,00,000 and Project B Rs. 8,00,000 declining at an annual rate of Rs.2,00,000.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. From the following particulars, prepare a Balance Sheet as on 31st March, 2011.

	Rs.		Rs.
Capital	50,000	Loan to Smith	5,000
Business Premises	55,000	Investments	3,000
Furniture & Fixtures	2,500	Cash in Hand	200
Bills Receivable	3,500	Cash at Bank	3,500
Bills Payable	2,500	Proprietor's Withdrawals	3,000
Sundry Debtors	20,000	Net Profit	38,900
Sundry Creditors	15,800	Closing Stock	7,000
Packing Machinery	4,500		

10. From the following balances extracted from S.K. Company Limited as on 31st December, 2009 and 2010, you are required to prepare Schedule of Changes in Working Capital and funds flow statement.

Liabilities	As on 31st December 2009 Rs.	2010 Rs.	Assets	As on 31st December 2009 Rs.	2010 Rs.
Share Capital	1,00,000	1,10,000	Building	40,000	38,000
General Reserve	14,000	18,000	Plant & Machinery	37,000	36,000
Profit & Loss Account	16,000	13,000	Long -term		
Creditors	8,000	5,400	Investment	10,000	21,000
Bills Payable	1,200	800	Stock	30,000	23,400
Provision for Tax	16,000	18,000	Bills Receivable	2,000	3,200
Provision for			Debtors	18,000	19,000
Doubtful Debts	400	600	Cash at Bank	6,600	15,200
			Preliminary Expenses	12,000	10,000
Total	1,55,600	1,65,800		1,55,600	1,65,800

Additional Information:

(a) Depreciation charged on Plant was Rs.4, 000.

(b) Provision for taxation Rs.19, 000 was made during the year 2010.

(c) Interim dividend of Rs. 8,000 was paid during the year.

(d) A Piece of machinery was sold for Rs.8, 000 during the year 2010. It had costed Rs.12, 000, depreciation of Rs.7, 000 has been provided on it.

11. What are the uses and limitations of break even chart? How you would locate the break even point on the chart?

12. The standard material required to manufacture one unit of Product X is 10 kg and the standard price per kg of material is Rs.2.50. The cost accounts records, however, reveal that 11,500 kg of materials costing Rs.27,600 were used for manufacturing 1,000 units of Product X. Calculate the material variances.

13. Prepare a Cash Budget for the three months ending 30th June, 1997 from the information given below :

(a)

Month	Sales (Rs.)	Materials (Rs.)	Wages (Rs.)	Overheads (Rs.)
February	14,000	9,600	3,000	1,700
March	15,000	9,000	3,000	1,900
April	16,000	9,200	3,200	2,000
May	17,000	10,000	3,600	2,200
June	18,000	10,400	4,000	2,300

(b) Credit terms are :

Sales/Debtors — 10% sales are on cash, 50% of the credit sales are colleted next month and the balance in the following month.

Creditors	Materials	2 months
	Wages	1/4 months
	Overheads	1/2 month

(c) Cash and Bank balance on 1st April 1997 is expected to be Rs.6,000.

(d) Other relevant information are:

(i) Plant and Machinery will be installed in February, 1997 at a cost of Rs. 96,000. The monthly installment of Rs. 2,000 is payable from April onward.

(ii) Dividend @ 5% on Preference Share Capital of Rs.2,00,000 will be paid on 1st June.

(iii) Advance to be received for sale of vehicles Rs.9,000 in June.

(iv) Dividends from investments amounting to Rs. 1,000 are expected to be received in June.

(v) Income tax (advance) to be paid in June, is Rs.2,000.

14. What factors would you take into consideration in planning the working capital requirement of a firm? Explain them in brief.

15. A limited company is considering the purchase of a machine. Two machines A and B are available. From the following information relating to the two machines, ascertain which of the two alternatives will be more profitable. The average rate of tax may be taken at 50%. Use average rate of return method.

	Machine A	Machine B
Cost of each machine	Rs. 50,000	Rs. 80,000
Working life	4 years	6 years
Earnings before tax:	Rs.	Rs.
Year I	10,000	8,000
Year II	15,000	14,000
Year III	20,000	25,000
Year IV	15,000	30,000
Year V	–	18,000
Year VI	–	13,000

DE–4174 **402**

DISTANCE EDUCATION

M.C.A. (S) /M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is the purpose of Encryption? Explain.

2. Define CGI. Draw flow chart of Java Data types.

3. Write a VB Script to find factorial of a number.

4. Explain, how to get data input from the user, using Java Script.

5. Differentiate the ordered and unordered Lists with suitable examples.

6. Explain HTML style sheet. What are the advantages of using it?

7. Discuss about the VDO live Technology.

8. How to access the multimedia components of animation and image in web pages?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the various states in multithread.

10. Briefly explain about Perl.

11. Explain the various built in objects of Java Script.

12. Write short notes on:

    (a) Internet Explorer

    (b) Shockware.

13. Define Active X control. Explain how it works?

14. Discuss in detail:

    (a) Dynamic Web pages

    (b) Netscape navigator plug-ins.

15. Create a simple web page with two frames. Provide necessary links between the pages. Your page should also insert image and Animation.

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DE–4175 **403**

DISTANCE EDUCATION

M.C.A (S)/M.C.A(S) Lateral DEGREE EXAMINATION, MAY 2012.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. What is object oriented analysis and design? Explain the principles involved in object oriented system development.

2. Discuss the importance of proper classification. Briefly explain the different approaches used for identifying classes and objects.

3. Explain refactoring.

4. Describe the basic activities of Object oriented analysis and explain how Use–Case modeling is useful in analysis.

5. Explain how objects are mapped with relational system (tables).

6. What is a Component? Explain how Component Diagrams are drawn with an example.

7. Write a note on abstractor.

8. Explain transition phase in the development process.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What are the various diagrams that are used in analysis and design steps of Booch Methodology? Explain with your own example.

10. Write notes on:

    (a) Association relationship.

    (b) Deployment diagram.

11. (a) What are the elements of a class diagram? Explain with example.

    (b) Discuss any two advanced concepts of class diagram and their significance.

12. Discuss about UML deployment and component diagrams. Draw the diagrams for a banking application.

13. Explain about implementation model (mapping design to code).

14. Explain with an example interaction diagram.

15. How does UML help in the Analysis and Design of OO systems?

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DE–4176 **404**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define Error Handling, code optimization and Regular expression.

2. Write short notes on Book Keeping.

3. Explain Handle pruning with an example.

4. Explain Right most Derivation and left most derivation tree with an example.

5. Write short notes on Three address code.

6. Discuss Semantic errors.

7. Explain briefly about Loop optimization.

8. Write a note on problems in code generation.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about the phases of a compiler. (15)

10. Explain Bottom-up parsing with an example. (15)

11. Explain how to construct SLR parsing tables. (15)

12. Write down the moves made by the Bottom up parser for the input 3*5 +4n in synthesized attributes. (15)

13. Explain various data structure for symbol table. (15)

14. Explain in detail:

    (a) Peephole optimization. (7)

    (b) Principle sources of optimization. (8)

15. Construct NFA and DFA for (aa*/bb*). (15)

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DE–4177 **501**

DISTANCE EDUCATION

M.C.A. (SEM)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

DISTRIBUTED COMPUTING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Define IP Address. Explain the various classes of IP Address with neat sketch.

2. Discuss the important characteristics of Inter process communication (IPC).

3. Write short notes on digital signatures.

4. Explain the design requirements for a system to synchronize the clocks in a distributed system.

5. Illustrate the concept of deadlock with neat sketch.

6. What is data gram? What are the important issues related to data gram? Discuss.

7. Explain about Remote procedure call (RPC).

8. Discuss the security policy and its mechanisms in the case of a multi-user operating system such as UNIX.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Describe the OSI model of network and compare the services provided by the layers with TCP/IP model.

10. Discuss in detail about the semantics and design a protocol for a group form of request-reply interaction.

11. Give a detailed note on any two cryptographic algorithms with suitable illustrations.

12. Write a brief note on Domain Name system.

13. Explain briefly about two phase commit protocols with neat sketch.

14. Discuss briefly about frame format, operation and different medium options of Ethernet with neat sketch.

15. Describe, why the algorithm for reliable multicast over IP multicast does not work for open groups. Justify your answer.

DE–4178 **502**

DISTANCE EDUCATION

M.C.A (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define the following HTML tags.

   (a) List (b) Tables (c) Images (d) Forms.

2. Define DOM and SAX using XML processors in detail.

3. List out the advantages of Java Beans.

4. What is Java Beans API? Explain in detail.

5. Describe the Java X. Servelet package with suitable examples.

6. Describe the anatomy of a JSP page and explain JSP processing with suitable example.

7. Describe Tomcat server and Testing Tomcat in detail.

8. Explain how to access a database from a JSP page with example.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is Java script? Explain Dynamic HTML with Java script in detail.

10. Explain HTML frames and cascading style sheets in detail.

11. Explain the following java beans concepts.

    (a) BDK (b) Introspection (c) Constrained properties.

12. What is servelets? Explain the life cycle of a servelets.

13. Explain the JSP application design with MVC setting up and JSP environment in detail.

14. Explain JSP application development in detail.

15. Explain database programming using JDBC with suitable code.

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DE–4179 **503**

DISTANCE EDUCATION

M.C.A. (S) /M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

.NET FRAME WORKS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain about common language runtime.

2. Briefly explain about common type system.

3. What is attribute? Explain how to construct an attribute.

4. How to create Multiple Document Interface Application?

5. Describe in detail about the lifecycle of web form.

6. Compare ASP with ASP. NET.

7. Describe the properties and methods of any five controls in windows forms.

8. Why we need .Net web services?

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain in detail about different .Net building blocks.

10. Describe in detail the reflection API.

11. What is thread? Explain the joining and killing threads with suitable example.

12. Write a program using ADO .NET to connect to the northwind database and read the name of the employee. The employee table has 2 field namely first name and last name.

13. Explain in detail about ADO .NET Data provider.

14. What are the task available in web service development?

15. Explain the SOAP architecture and its futures.

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DE–4180 **504**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

DATA MINING AND WAREHOUSING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a short note on Data mining in CRM.

2. Write about

   (a) Classification Accuracy

   (b) Transparency

   (c) Statistical significance

   (d) Information content.

3. Explain the concepts of coding in KDD process.

4. How query tools can be used for preliminary analysis of the data set?

5. Explain about Extract and load process of a Data Warehouse.

6. What is an Aggregation? Explain.

7. Write a short note on Security issues in Data warehouse.

8. What are AdHoc Queries?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the machine learning and methodology of science with an example.

10. What is a data warehouse? Why do we need it?

11. Explain the usage of Decision trees in knowledge discovery process.

12. Explain the concepts of data marting in detail.

13. Explain the role of data warehousing process managers.

14. Give the importance of back-up and recovery in data warehouse.

15. Explain in detail about capacity planning.

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DEΘ4181 **601**

DISTANCE EDUCATION

M.C.A(S)/M.C.A(S)Lateral DEGREE EXAMINATION,
MAY 2012.

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Acadamic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define CDMA.

2. What is meant by localization ? Explain briefly.

3. Explain about the different types of WLAN.

4. Write short notes on blue tooth.

5. What do you mean by Radio Network controller?

6. Explain about the components of mobile IP.

7. Explain the features of Mobile TCP.

8. Explain about Datagram protocol.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Describe in detail about the principles of digital audio and video broad casting.

10. Explain in detail about the architecture of UMTS.

11. Write notes on tunneling and encapsulation.

12. Explain with neat diagram the working principle of Wireless Application Environment (WAE).

13. Explain the components of GSM in detail.

14. Write notes on Wireless Telephony Application.

15. Write short notes on

    (a) Minimal Uncapsulatio

    (b) Route optimization.

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DE–4182 **602**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Write a brief note on MVC architecture.

2. Enumerate the features of 3 tier client server architecture model.

3. Explain the salient features of CORBA.

4. List the different servlet API features.

5. Compare stateful and stateless session beans.

6. Write a brief note on ASP.NET page structure.

7. Write a brief note on Adrotator controls used in ASP.NET.

8. List the steps involved in connecting a webservice to a data source.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Explain the J2EE architecture.

10. (a) Write a brief note on JDBC. (10)

    (b) Write a sample code used to connect a Database using JDBC. (5)

11. What is EJB? Explain the EJB component model.

12. Write a Stateless Session Bean Which Coverts the Dollar into Rupees and Rupees Into Dollar.

13. Explain the different Web server controls available in ASP.NET with their properties.

14. Explain the use of WSDL and SOAP in describing web services.

15. Design a simple website using ASP.NET to display the courses available in a college.

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DE–4183 **603**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART — I ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Explain the History of Artificial Intelligence.

2. Give a brief note on Game Playing.

3. Explain the First order logic Inference.

4. Explain the Unification Algorithm.

5. What is the signification of conditional Planning.

6. Write short notes on Planning and Acting in Non-Deterministic domain.

7. Give a brief note on the Axioms of probability.

8. What are the different types of knowledge Representation techniques?

PART — II ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Give a detailed distribution on the structure of Agents.

10. Explain the concept of problem solving with an example.

11. Bring out the significance of forward chaining.

12. Explain objects, Actions and events with suitable diagrams.

13. Give a detailed note on the planning problem.

14. Explain the sequential Decision Problem.

15. List the various steps involved in analysing a sentence in Natural Language Processing.

DE–4184 **604**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Discuss Foss.

2. Explain system environment.

3. What is process in LINUX? Explain.

4. Write a note on X-Window system.

5. What is CVS? List some useful CVS commands.

6. What is the benefit of the –q option of ar?

7. Write a note on Bash.

8. What are the roles for local and global variables in python?

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain services in LINUX. How services are selected for stertup in Linux.

10. Explain computation in shell programming with example.

11. Write short notes on the following :

    (a) File compression

    (b) Web browsers

    (c) E-mail services.

12. Explain the process of setting a firewall using net finder, and IP tables.

13. Explain the usage of CVS tool to manage source code revision.

14. Discuss about the common programming practices and guidelines for GNU/LINUX.

15. Explain about X-Windows server architecture in detail.

DE–4403 **205**

DISTANCE EDUCATION

M.C.A. (Semester)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

Second Semester

C++ AND GRAPHICS LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks:

Records : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Examiners has to choose ONE question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Develop a C++ program to draw a Triangle using Bresunham's algorithm.

   (b) Develop a C++ program to draw a square and fill it with eight colors equally.

---------- Cut here ----------

2. (a) Write a C++ program to draw a straight line using simple DDA technique.

   (b) Draw a square translate it to a distance and then rotate it through an angle Anti-clock wise.

3. (a) Write a C++ program to create a ball and make the ball move on the sine curve path.

(b) Write a C++ program to draw 3-D cube and perform animation on it.

---------- Cut here ----------

4. (a) Write a C++ program to draw the square fill at with two colors equally.

(b) Develop a C++ program to draw rectangle divide the object in to four quadrants and fill each quadrant with color.

---------- Cut here ----------

5. (a) Develop a C++ program to draw a triangle divide the object in to three equal parts fill each part with different colors.

(b) Write a C++ program to draw a ellipse with Bresunham's algorithms.

---------- Cut here ----------

6. (a) Write a C++ program to implement curve clipping algorithm using graphic functions.

(b) Write a procedure and C++ program to draw linear object.

---------- Cut here ----------

7. (a) Write a C++ program to draw non-linear object curve.

(b) Develop a C++ program to draw non-linear object ellipse.

8. (a) Write a C++ program to manipulate the circle graphically using shading technique.

(b) Develop a C++ program to draw rectangle using simple DDA technique.

---------- Cut here ----------

9. (a) Write a C++ program to show graphical transformation translation of two dimentional object.

(b) Implement the polygon clipping algorithm using C++ graphics mode graphic functions.

---------- Cut here ----------

10. (a) Write a C++ program to draw a sphere and perform animation on it.

(b) Draw ellipse using 3-D object and animation on it.

DE–4404 **206**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2012.

Second Semester

UNIX AND SHELL PROGRAMMING LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Each candidate should select ONE question using lot system.

Each subdivision carries 50 marks.

1. (a) Write a shell program for the following :

   (i) Enter any ten numbers into the file.

   (ii) Sort them in ascending order.

   (b) Write a shell program to concatenate and copy two strings without using built – in functions.

---------- Cut here ----------

2. (a) Write a menu driven shell program for the following :

   (i) Display the current date and time of the system.

   (ii) Display the current directory.

   (b) Write a shell program to reverse the given string using built – in functions.

3. (a) Write a shell program to display the natural numbers between 1 ..... 100.

(b) Write a menu driven shell program to perform the following :

(i) Copy a file.

(ii) Edit a file.

(iii) Delete a file.

---------- Cut here ----------

4. (a) Write a shell program to find the factorial of given 'n' numbers.

(b) Write a shell program to perform the following :

(i) Read any text file.

(ii) Print its contents without using 'cat'/'pr' commands.

---------- Cut here ----------

5. (a) Write a shell program to calculate the fibonacci series between 1 ..... 100.

(b) Write a shell program to create a text file with any ten flower names and also perform the following.

(i) Open the file in write mode.

(ii) Arrange the words in ascending order.

6. (a) Write a shell program to accept the file as an input and perform the following :

   (i) Count and display the number of lines in that file.

   (ii) Print lines not matching the given pattern.

   (b) Write a shell program to count the digits of a given integer without using built – in functions.

---------- Cut here ----------

7. (a) Write a shell program to sum up the series $1^2 + 3^2 + 5^2 + 7^2 + ... + 99^2$.

   (b) Write a shell program to display the prime numbers between 1 ..... 100.

DE–4405 **305**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

NETWORK LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Each candidate should select ONE question using lot system.

Each subdivision carries 50 marks.

1. (a) Write a Java program to add two numbers using RMI client, server technique.

   (b) Write a Java program to send message from one system to another system.

---------- Cut here ----------

2. (a) Write a Java program to find factorial of given number using RMI client, Server technique.

   (b) Write a Java program to read file from the remote system.

---------- Cut here ----------

3. (a) Write a Java program to add two numbers using RMI client, server technique.

   (b) Write a Java program to find factorial of given number using FACTORIAL INTERFACE technique.

4. (a) Write a Java program to find factorial of given number using RMI client, Server technique.

(b) Write a Java program to setup a chat server and chat client.

---------- Cut here ----------

5. (a) Write a Java program to find factorial of given number using FACTORIAL INTERFACE technique.

(b) Write a Java program to setup echo server and echo client.

---------- Cut here ----------

6. (a) Write a Java program to setup a chat server and chat client.

(b) Write a Java program to display the user and their IP addresses logged in the network.

---------- Cut here ----------

7. (a) Write a Java program to setup echo server and echo client.

(b) Write a Java program to ENCRYPT given word.

---------- Cut here ----------

8. (a) Write a Java program to display the user and their IP addresses logged in the network.

(b) Write a Java program to setup Time server and Time client.

DE–4406 **306**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

VISUAL PROGRAMMING LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by the lot system.

1. (a) Write a VB program to read a year in a text box and to print whether it is a leap year or not. (40)

   (b) Write a Visual C++ program to create a list box and to displays name of the states using MFC. (60)

---------- Cut here ----------

2. (a) Write a VB program to create a application to insert, update, delete and retrieve records from a database. (60)

   (b) Write a Visual C++ program to create a new font using MFC. (40)

---------- Cut here ----------

3. (a) Write a event procedure to perform the following

   (i) Reverse a string

   (ii) To check whether a string is a palindrome or not. (50)

   (b) Write a Visual C++ program to displays current mouse coordinates in status bar. (50)

4. (a) Write a VB program to simulate a simple calculator with memory feature (memory in and memory recall). (50)

(b) Write a visual C++ program to create a window of desired size using MFC. (50)

---------- Cut here ----------

5. (a) Write a VB program to read two numbers from text boxes and to list all numbers which are divisible by 5 and between the two numbers read. (40)

(b) Write a Visual C++ program for Mouse Events. (60)

DE–4407 **405**

DISTANCE EDUCATION

M.C.A. (Computer Applications) M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

INTERNET PROGRAMMING LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a Java Script to create a color palet and display the text in the color chosen from the palet with proper background color.

   (b) Write a Java program using Applet to display the dialogue menu in applet.

---------- Cut here ----------

2. (a) Write a VB Script to do the following :

   (i) Check the given password

   (ii) Change the existing password.

   (b) Write a Swing program to create Buttons with

   (i) Tool tip text

   (ii) Image

   (iii) Border

   (iv) Short cut key.

3. (a) Develop a web page for job recruitment agency in an IT industry.

(b) Write a Java Script to create a window by using the confirm message.

---------- Cut here ----------

4. (a) Using atleast6 20 HTML tags, create a screen with a string "WEB DESIGN".

(b) Write a Java program using applet to display any three images when three buttons in the border layout are clicked.

---------- Cut here ----------

5. (a) Write a swing program to create the Tabbed panels.

(b) Write a VB Script to create a calendar for given month and year.

DE–4408 **406**

DISTANCE EDUCATION

M.C.A. (Computer Applications)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

COMPILER DESIGN – LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner should select and give ONE question to each candidate by lot system.

---------- Cut here ----------

1. (a) Write a C program to recognize the integer constant in the given source code.

   (b) Write a C program to convert the infix expression $a + b * c - d$ into postfix.

2. (a) To draw the transition diagrams for DFA by accepting the following regular expression

(i) 12/53

(ii) $(a/b)^* abb$.

(b) Write a C program to remove comment lines from the given source code.

---------- Cut here ----------

3. (a) Write a C program to accept a string and check whether it satisfies the following production rule by using operation precedence grammar.

$S \rightarrow BaC$

$B \rightarrow bCa$

$C \rightarrow b/Cab$.

(b) Write a C program to recognize the keywords in the given source code.

---------- Cut here ----------

4. (a) Write a C program to construct the right most derivations to derive a word $(a,(a,a))$ using the grammar

$S \rightarrow (T)$

$S \rightarrow a$

$T \rightarrow S$

$T \rightarrow T,S$.

(b) Write a C program for DFA to accept the regular expression $(a/b)^*$.

5. (a) Write a C program to produce line members in the given source code.

(b) Write a C program to construct of an SLR parsing table for the following grammar.

$E \rightarrow E + T / T$

$T \rightarrow T * F / F$

$F \rightarrow (E) / id$.

---------- Cut here ----------

6. (a) Write a C program to construct predictive parsing table for the following production rule

$S \rightarrow iCtSS1 / a$

$S1 \rightarrow eS / E$ (F as empty)

$C \rightarrow b$.

(b) Write a C program to convert the given infix $(a + b * c) - (c * b)$ into postfix.

---------- Cut here ----------

7. (a) Write a C program to translate the expression $(A + B) * (C + D)$ into Quadruples, Triples and Indirect triples.

(b) Write a C program to recognize the identifier in the given source code.

8. (a) Write a C program to construct of an canonical LR parsing table for the following grammar

$S \rightarrow AA$

$A \rightarrow Aa/b$.

(b) Write a C program for DFA to accept the regular expression $(a/b)(a/b)$.

DE–4409 **505**

DISTANCE EDUCATION

M.C.A. (Computer Applications)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

.NET LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

---------- Cut here ----------

1. (a) To create employee table and insert a few records using disconnected accessing using VB.NET.

   (b) Write a program to generate the factorial operation using ASP.NET.

---------- Cut here ----------

2. (a) To develop a pay roll project to updata and delete few records using disconnected access using VB.NET.

   (b) Design a simple website that makes use of master pages using ASP.NET.

3. (a) Develop a web service to fetch a data from a table and send it across to the client using ASP.NET.

(b) To develop a student information system project to view the records using Grid view in VB.NET.

---------- Cut here ----------

4. (a) To develop a student information system project to view the records using details view in VB.NET.

(b) To design a books details webpage that make use of Ad Rotator control using ASP.NET.

---------- Cut here ----------

5. (a) To develop an inventory control project to generate a crystal report from an existing database using VB.NET key in, Debug and execute program.

(b) Write a program to perform money conversion using ASP.NET.

---------- Cut here ----------

6. (a) Write a program to implement paging concept data grid and data set using ASP.NET.

(b) Write a program in VB.NET to create a Notepad.

---------- Cut here ----------

7. (a) To design a webpage involving multi view or wizard control using ASP.NET.

(b) Write a program in VB.NET to create a calculator.

8. (a) Design an Auto mobile shop webpage involving two hotspots in a webpage using ASP.NET.

(b) To develop a inventory control project to generate a crystal report from an existing database using VB.NET key in, Debug and execute program.

DE–4410 **506**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

Computer Applications

WEB TECHNOLOGY LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Create an HTML Form Design of employees information.

(b) Create a web page in the format of front page of a mobile show room, various mobile features and prices using text links. Align the text with colors.

2. (a) Write an XML program which Displays the breakfast menu at Hotel. Format the document with XSL stylesheet.

(b) Write an ASP program to calculate the electricity bill using Java Script

Note :

No. of units	Amount
< = 100	1 Re.
> 100 and < 200	2 Rs.
> 300	4 Rs.

---------- cut here ----------

3. (a) Create an XML document whose root element is movies. Its child elements are movie, title, writer, producer, Director, Author, comments. Creating a cascading style sheet.

(b) Write an ASP program to prepare employee pay bill using Javascript.

---------- cut here ----------

4. (a) Develop a servlet program to display your favourite writer.

(b) Write a JSP program to calculate student mark list using IDBC.

---------- cut here ----------

5. (a) Write a servlet program to send email message.

(b) Develop a javascript for animating an image.

---------- cut here ----------

6. (a) Create a HTML Frame Design using Text links. Align the text with color.

(b) Create a webpage in the format of front page of news paper using text links. Align the text with various color.

7. (a) Design a HTML form using various controls.

(b) Write a JSP program to display Book-id, Book-name, publisher-name. Author-name, price, ISBN-No. Using IDBC.

---------- cut here ----------

8. (a) Write an XML Document to Display your favourite Indian Music Director. Write an XSL style sheet and attach that to the XML document. Validate the document using DTD or XSD.

(b) Write an JSP program to prepare the Invoice Bill using Javascript.

DE–4411 **605**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

OPEN SOURCE PROGRAMMING LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

---------- Cut here ----------

1. (a) Create an PHP program for a school home page with different lines for various pages.

   (b) Write a shell program to find the biggest of four given numbers.

---------- Cut here ----------

2. (a) Create a PHP program for your class. With a group of your class on the home page. Have lines to the webpage containing the following information.

   (i) Monthly mark list of all students

   (ii) Class time table

   (iii) Students achievement

   (iv) Announcement.

   (b) Write a shell program to find the factorial of a given number.

3. (a) Write a PHP program for railway reservation assume your own data.

(b) Write a shell program to find the average of N numbers.

---------- Cut here ----------

4. (a) Write a program for income tax calculator with three different slabs.

(b) Write a shell program to find the area of circle.

---------- Cut here ----------

5. (a) Write a PHP program for a paybill calculation with deduction and netpay.

(b) Write a shell program for binary search.

---------- Cut here ----------

6. (a) Write a PHP program for an online shopping.

(b) Write a shell program for converting binary numbers to decimal.

---------- Cut here ----------

7. (a) Create a PHP program to create your resume and store it in a table.

(b) Write a shell program to find whether the given year is leap or not.

8. (a) Create a PHP program to create a radio group with three buttons 1, 2, 3. Depending on the choice of the user a separate message should be displayed for each of the radio buttons.

(b) Write a shell program for generating fibonocii series.

DE–4412 **606**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

MULTIMEDIA TOOLS LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-Up Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Programs	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Create a flash movie for object movement using Motion Tweening.

   (b) Implement a program to create tree using rendering.

---------- Cut here ----------

2. (a) Create a flash movie to add motion guide.

   (b) Generate a photoshop image, to display filter effects.

3. (a) Illustrate a flash movie to display the masking effects.

(b) Develop a web page using dream viewer for presentation of demo.

---------- Cut here ----------

4. (a) Create a flash movie to display cell animation.

(b) Create a photoshop image, which displays the feather effect.

---------- Cut here ----------

5. (a) Create a flash movie that performs the shape tweening.

(b) Create a photoshop image, to show layer style and blending options.

---------- Cut here ----------

6. (a) Create a flash movie for cloudy effects with rainfall.

(b) Implement a Maya program to advertise the pen using rendering.

---------- Cut here ----------

7. (a) Create a flash movie to blur the image and shadow effect of the same image.

(b) Create a movie with lighting and without lighting in Maya.

---------- Cut here ----------

8. (a) Create a photoshop image to display raster-images and vector images.

(b) Animate your own text with different colors and shading. Make the characters to display in different angles.